AI 500 - Foundations of AI

Quiz 4 - KDE, GMM, Hierarchical Clustering (A) - 18th October 2025

Name:_____ Roll Number:_____

Please note that MCQs can have multiple correct options.

Q1. Which of the following are true for hierarchical clustering? **(1 mark)**

A. **The method can uncover nested structure within data even without specifying k in advance.**
B. The "rigidity" of hierarchical clustering implies more adaptability to new data compared to iterative methods like K-means.
C. **It can be built top-down or bottom-up.**
D. Hierarchical clustering always guarantees the globally optimal set of clusters for a given dataset.
E. All of the above

Q2. Which of the following statements about divisive hierarchical clustering is correct? **(1 mark)**

A. It begins with each data point as its own cluster and repeatedly merges the closest pairs.
B. **It begins with all data points in a single large cluster and recursively splits it into smaller clusters.**
C. It produces results identical to agglomerative clustering for any given dataset.
D. Each split increases the number of clusters by two, and the process can be visualized as a dendrogram showing successive divisions.
E. All of the above

Q3. In a scenario where sensors record temperature readings at different spatial locations, which statement best describes the primary advantage(s) of using Kernel Density Estimation (KDE) for this application? **(1 mark)**

A. **KDE can interpolate values at unobserved locations by creating a continuous density estimate**
B. KDE automatically identifies optimal sensor placement locations
C. **KDE provides a non-parametric approach that doesn't assume a specific distribution (like Gaussian) for the data**
D. KDE can only be applied to one-dimensional data
E. All of the above

Q4. Consider a 2-component GMM where both components have equal mixing weights ($\pi_1 = \pi_2 = 0.5$) but different parameters: $\mu_1 = 50$, $\sigma_1 = 5$ and $\mu_2 = 150$, $\sigma_2 = 5$. For a data point $x = 52$, which statement correctly describes the contribution of each component to the likelihood? **(1 mark)**

A. Both components contribute equally to $p(x = 52)$ because the mixing weights are equal

B. Component 1 contributes more to p(x = 52) because the Gaussian density $g(52|\mu_1, \sigma_1)$ is much higher than $g(52|\mu_2, \sigma_2)$

C. The contribution of component k to the total likelihood is $\pi_k \cdot g(x|\mu_k, \sigma_k)$, which depends on both the mixing weight and the Gaussian density

D. Equal mixing weights guarantee that the responsibilities $P_1(52)$ and $P_2(52)$ will be equal

E. All of the Above

Q5. In a Gaussian Mixture Model (GMM) fitted using the Expectation-Maximization (EM) algorithm, the initial parameter settings and convergence properties significantly influence the outcome. Which of the following statements about parameter initialization and convergence in a GMM are correct? **(1 mark)**

A. Random initialization of means may result in convergence to different local maxima of the likelihood, often necessitating multiple restarts to approximate the global maximum with higher probability.

B. The EM algorithm's convergence to a high-likelihood solution is assured if the initial mixing weights are set equal (e.g., $\pi_k = 1/K$), regardless of the initial means and variances.

C. Initializing means with K-means clustering can enhance convergence stability by providing initial cluster centers that reflect data structure, though it may still lead to suboptimal local maxima if the data has overlapping clusters.

D. The EM algorithm's convergence speed is primarily determined by the initial parameter proximity to the true parameters, ensuring faster convergence when starting near the optimal solution.

E. All of the above

Q3. You are given the following four 2D data points:

- (0,0)
- (1,0)
- (0,1)
- (2,2)

For bandwidth *h=1*, use the *Epanechnikov kernel* to estimate the density at x = (0.5,0.5)

Epanechnikov kernel:

$$K(\mathbf{u}) = \begin{cases} \frac{3}{4}(1 - \|\mathbf{u}\|^2), & \text{if } \|\mathbf{u}\| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Answer:

$$\hat{p}(\mathbf{x}) = \frac{1}{4} \sum_{i=1}^{4} K(\mathbf{x} - \mathbf{x_i})$$

Point 1 distance: $\sqrt{(0.5 - 0)^2 + (0.5 - 0)^2} = 0.707$

Point 2 distance: $\sqrt{(0.5 - 1)^2 + (0.5 - 0)^2} = 0.707$

Point 3 distance: $\sqrt{(0.5-0)^2+(0.5-1)^2}=0.707$

Point 4 distance: $\sqrt{(0.5-2)^2+(0.5-2)^2}=2.121$ (outside kernel, will be 0)

(2 marks for correct distance calculation)

$\widehat{p}((0.5,0.5))=\frac{1}{4}(\frac{3}{4}(1-0.707^2)+\frac{3}{4}(1-0.707^2)+\frac{3}{4}(1-0.707^2)+0)=0.28125$

(2 marks for correct input into the formula)

(1 mark for correct answer)

AI 500 - Foundations of AI

Quiz 4 - KDE, GMM, Hierarchical Clustering (B) - 18th October 2025

Name:_____ Roll Number:_____

Please note that MCQs can have multiple correct options.

Q1. Which of the following statements about agglomerative hierarchical clustering is correct? **(1 mark)**

A. It begins with all data points in a single large cluster and recursively splits it into smaller clusters.
B. It begins with each data point as its own cluster and repeatedly merges the most similar clusters.
C. It guarantees the same final clusters regardless of the linkage method used.
D. Each merge reduces the number of clusters by half, and the process can be visualized as a dendrogram showing successive merges.

Q2. Which of the following are false for hierarchical clustering? **(1 mark)**

A. It can be built top-down or bottom-up.
B. The method can uncover nested structure within data even without specifying k in advance.
C. Hierarchical clustering always guarantees the globally optimal set of clusters for a given dataset.
D. The "rigidity" of hierarchical clustering implies more adaptability to new data compared to iterative methods like K-means.

Q3. In a GMM with K components, the mixing weights π_k must satisfy certain constraints. Which of the following statements about mixing weights are mathematically correct? **(1 mark)**

A. The sum of all mixing weights must equal 1: $\sum_k \pi_k = 1$, making them form a probability distribution over components
B. Mixing weights are updated in the E-step of the EM algorithm based on the responsibilities computed for each data point
C. The mixing weights represent the prior probability that a randomly selected data point comes from component k
D. The mixing weight π□ must always be less than 1/K to ensure proper normalization across K components

Q4. Consider a 2-component Gaussian Mixture Model (GMM) with mixing weights $\pi_1 = 0.7$ and $\pi_2 = 0.3$, and parameters $\mu_1 = 100$, $\sigma_1 = 10$ and $\mu_2 = 180$, $\sigma_2 = 10$. For a data point x = 110, which of the following statements correctly describe the contribution of each component to the likelihood p(x = 110)?

A. Component 1 contributes more to p(x = 110) because both its mixing weight is higher AND x = 110 is closer to $\mu_1 = 100$
B. Component 1 contributes more to p(x = 110) solely because it has a higher mixing weight ($\pi_1 > \pi_2$)
C. The contribution of component k to the total likelihood is given by $L_k(x) = \pi_k \cdot g(x|\mu_k, \sigma_k)$, which depends on both the mixing weight and the Gaussian density
D. The responsibility $P_1(110)$ will be greater than 0.7 because Component 1 benefits from both a higher mixing weight and the proximity of x = 110 to $\mu_1 = 100$
E. All of the above

Q5. In a scenario where sensors record temperature readings at different spatial locations, which statement best describes the primary advantage(s) of using Kernel Density Estimation (KDE) for this application? **(1 mark)**

A. KDE can only estimate the probability density of temperature values, not predict actual temperatures at new locations
B. KDE automatically determines the optimal number of sensors needed for accurate temperature mapping
C. KDE eliminates the bandwidth selection problem by using the spatial distances between sensors to set kernel widths
D. KDE can predict temperature values at unsampled locations by providing a smooth, continuous density estimate across the spatial domain
E. All of the Above

Q3. You are given the following four 2D data points: **(5 marks)**

- (0,0)
- (1,0)
- (0,1)
- (2,2)

For bandwidth *h=1*, use the *Epanechnikov kernel* to estimate the density at x = (1.0,1.0)
Epanechnikov kernel:

$$K(\mathbf{u}) = \begin{cases} \frac{3}{4}(1 - \|\mathbf{u}\|^2), & \text{if } \|\mathbf{u}\| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Answer:

$$\hat{p}(\mathbf{x}) = \frac{1}{4} \sum_{i=1}^{4} K(\mathbf{x} - \mathbf{x_i})$$

Point 1 distance: $\sqrt{(1-0)^2 + (1-0)^2} = 1.414$ (outside kernel, will be 0)

Point 2 distance: $\sqrt{(1-1)^2 + (1-0)^2} = 1$

Point 3 distance: $\sqrt{(1-0)^2+(1-1)^2}=1$

Point 4 distance: $\sqrt{(1-2)^2+(1-2)^2}=1.414$ (outside kernel, will be 0)

(**2 marks for correct distance calculation)**

$\widehat{p}((1,1))=\frac{1}{4}(0+\frac{3}{4}(1-1^2)+\frac{3}{4}(1-1^2)+0)=0$

(2 marks for correct input into the formula)

(1 mark for correct answer)